

Hamzah Bany Ata

Robotics & Autonomous Systems Engineer

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Professional Profile

Robotics and Mechatronics Engineer specializing in the design, integration, and field deployment of autonomous ground and aerial systems. Over five years of experience spanning tactical UAV operations, field deployment, and needs-based R&D for military autonomous systems. Currently serving as a UAV R&D Engineer for the Jordanian Armed Forces, with extensive operational experience across military-grade autonomous platforms including Quantum Systems Vector, AeroVironment PUMA 3 AE, and tactical ISR VTOLs. Seeking a fully funded PhD position or a robotics R&D role focused on autonomous field robotic systems — specifically the design and integration of heterogeneous aerial-ground platforms with task-specific payloads for reliable deployment in unstructured outdoor environments.

Education

M.Sc. in Mechanical Engineering — Mechatronics Sep 2020 – May 2026

Jordan University of Science and Technology (JUST), Irbid, Jordan

GPA: 3.63 / 4.0

- **Note:** Studies completed concurrently with full-time military engineering service (2022-2026).
- **Thesis:** Development of a Landmine Detection and Removal System Integrated with the Jackal Robot — engineered a pneumatic detect-to-extract payload for the Clearpath Jackal integrating RTK/IMU navigation, analyzed through pilot field trials bounding geotechnical force and geometric clearance limits.

B.Sc. in Mechanical Engineering

Sep 2015 – Feb 2020

Jordan University of Science and Technology (JUST), Irbid, Jordan

GPA: 3.72 / 4.0 (Top 5%)

- **Graduation Project:** Educational kit for mass-spring-damper vibration systems; vibration measured via image processing and integrated with LabVIEW.

Publications

1. Bany-Ata, A., **Bany-Ata, H.**, Kokash, H., Baqain, S., & Shakoore, M. (2026). Multi-objective optimisation and exergoeconomic analysis of ammonia-based organic Rankine cycles for waste heat recovery in a Power-to-Ammonia-to-Power system. *Energy Conversion and Management*, 365, 121738. [Q1 Journal, Elsevier; 2nd author] — DOI: 10.1016/j.enconman.2026.121738
2. Bany-Ata, A., **Bany-Ata, H.**, Kokash, H., Baqain, S., & Shakoore, M. (2026). Dynamic Exergoenvironmental Priority Inversion in Power-to-Ammonia-to-Power Systems: The Static T0 Fallacy in Hot-Arid Climates. *Clean Technologies* (MDPI). [Q2 Journal, MDPI; 2nd author] — DOI: 10.3390/cleantechnol8040120

Relevant Experience

Jordanian Armed Forces — Drones Department

Aug 2022 – Present

Mechanical Engineer (Mechatronics) — UAV Research & Development

Amman, Jordan

- Engineered a data pipeline to communicate with and parse KLV metadata from tactical-grade payloads (e.g., Q40TIRM-HD). Explored NATO MISB ST 0601 standards to structure the UAS datalink local set and convert target geolocations to precise UTM coordinates; **received a special recognition and financial award from the Army Commander.**
- Developing a vision-based navigation architecture for autonomous UAV flight in GNSS-denied environments without loiter guidance, matching downward-facing camera feeds with pre-existing aerial imagery via a Jetson Orin Nano companion computer.
- Serve as a technical evaluator for VTOL, multi-rotor, and fixed-wing platforms running closed-source stacks, ArduPilot, and PX4, analyzing flight-controller autonomy behavior in unstructured field conditions.
- Manage tactical field deployments and hardware failures, including operating UAS platforms (Puma 3 AE, Quantum Systems Vector) in GNSS-denied environments utilizing camera guidance/ loiter flight mode.
- Direct field testing and deployment; serve as an advanced operator and technical evaluator for military UAS.
- Apply digital manufacturing (3D printing, CNC machining) to fabricate custom payload mechanisms.

Center of Excellence for Innovative Projects (CEIP) | JUST

Feb 2025 – Feb 2026

Student Project | System Integration

Irbid, Jordan

- Collaboratively built two heavy-lift multi-rotor platforms (foldable coaxial and rigid H-frame), leading systems integration, custom BMS design, and avionics.

Jordan University of Science and Technology (JUST)

Sep 2021 – Aug 2022

Research & Teaching Assistant

Irbid, Jordan

- Contributed to “Multi-Sensor Cooperative Robots for Shallow Buried Explosive Threat Detection,” funded by NATO’s Science for Peace Programme.
- Delivered 4 intensive workshops on MATLAB and Arduino fundamentals to 80+ undergraduate students.

Freelance Developer

Feb 2020 – Aug 2022

Self-employed

Remote

- Delivered 20+ projects in Arduino, simulation, and algorithm implementation for academic clients, including MATLAB-based medical image fusion and RRR manipulator kinematics.

UAS Operations & Certifications

Extensive field experience in deployment, mission planning, and operation of tactical autonomous systems.

- **Quantum Systems Vector™**: Certified UAS-Operator (Feb 2025). Qualified for GNSS-denied flight, convoy operations, and sensor guidance.
- **AeroVironment PUMA™ 3 AE**: Certified SUAS Operator (Jun 2024) and certified **Train-The-Trainer (TTT)** (Jun 2024), qualified to instruct other operators.
- **Military ISR VTOL**: Certified fixed-wing UAV operator (Jordan Armed Forces, 2025).
- **Commercial & FPV Fleets**: Extensive operational testing with DJI systems (Matrice 350 RTK, Matrice 4T, Mavic Enterprise) and 5/7/10-inch FPV platforms.

Technical Skills

Programming & Software: Python, C/C++, MATLAB, ROS/ROS2, OpenCV, Simulink

UAV Systems & Stacks: ArduPilot, PX4, Betaflight, RTK/IMU integration, KLV/MISB Standards

Hardware & Embedded: Jetson Orin Nano, ESP32, Battery Management Systems (BMS), CAN bus, I2C

Control & Algorithms: Kalman filtering, PID control, State-space modeling, Search & Optimization (A*, NSGA-II, ACO), Image Processing

Mechanical & Fabrication: SolidWorks, AutoCAD, Pneumatic Systems, 3D Printing, CNC Machining, PCB Milling

Awards

- **Army Commander Award** (2026) — for the real-time KLV/UTM coordinate transformation system.
- **2nd Place**, Projects Competition, Engineering College Scientific Day, JUST (2019).